

Reference

Bayes' rule	$P(A \mid B) = P(B \mid A) P(A) / P(B)$
Total probability	$P(B) = \sum_i P(B \mid A_i) P(A_i)$
Binomial	$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Linear classifier	$\hat{y} = \mathbb{I}[w^\top x + b > 0]$, boundary $w^\top x + b = 0$
Sigmoid	$\sigma(z) = 1 / (1 + e^{-z})$, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Scratch work (not graded)